

Kalman Filter

Grundlagen

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Table of contents

- 1 Introduction
- 2 Foundations
- 3 Simplified explanation
- 4 The Math behind
- 5 Practical example
- 6 Summary

Rudolf Emil Kalman

- was born in 1930 in Hungary
- Bachelor of Science and Master of Science at the MIT
- PHD 1957 at the Columbia
- developed the filter around the years 1960 to 1961
- died 2016 in Florida



Fig.: Rudolf Emil Kalman

Overview:

- Developed by Norbert Wiener in the 1940s.
- Minimizes mean square error between estimated and true signal.

Mathematical Foundation:

$$H(f) = \frac{S_{dx}(f)}{S_{xx}(f)}$$

- $S_{dx}(f)$: Cross-power spectral density.
- $S_{xx}(f)$: Power spectral density.

Applications:

- Noise reduction, image restoration, data smoothing.

Overview:

- Developed by Rudolf E. Kalman in the 1960s.
- Extends Wiener filter principles to time-variant systems.

Key Features:

- Handles non-stationary processes.
- Uses state-space representation.
- Operates recursively for real-time applications.
- Models process and measurement noise.

Prediction Step:

$$\hat{x}_{k|k-1} = F_{k-1}\hat{x}_{k-1|k-1} + B_{k-1}u_{k-1}$$

$$P_{k|k-1} = F_{k-1}P_{k-1|k-1}F_{k-1}^T + Q_{k-1}$$

Update Step:

$$K_k = P_{k|k-1}H_k^T(H_kP_{k|k-1}H_k^T + R_k)^{-1}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(z_k - H_k\hat{x}_{k|k-1})$$

$$P_{k|k} = (I - K_kH_k)P_{k|k-1}$$

Conclusion

- Wiener filter laid the groundwork for statistical signal processing.
- Kalman filter extended these principles for dynamic systems.
- Widely used in aerospace, robotics, finance, and more.

What is the Kalman Filter?

The Kalman filter is a mathematical method for iteratively estimating parameters to describe system states.

In this process, a prediction about a parameter value is repeatedly made, combined with the error-prone measurement, and then used again to make a new prediction.

Kalman Filter

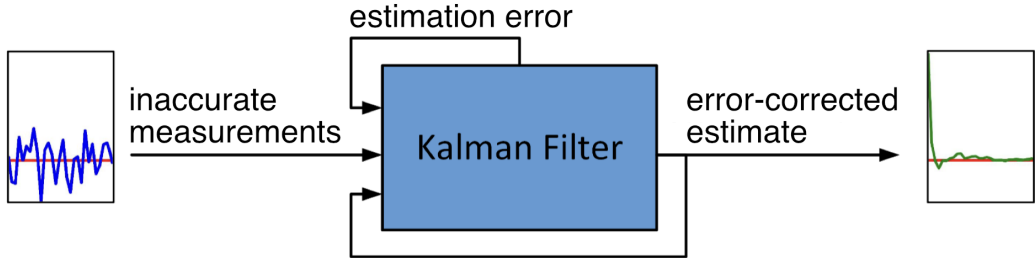


Fig.: Procedure of the Kalman Filter

Sensors susceptible to failure

- Light sensor (photoresistor)
- Ultrasonic sensor
- Infrared sensor
- Temperature sensor (thermoelectric)
- Magnetic field sensor
- Gas and air quality sensor
- Humidity sensor
- Motion sensor (accelerometer)
- Pressure sensor
- Sound sensor

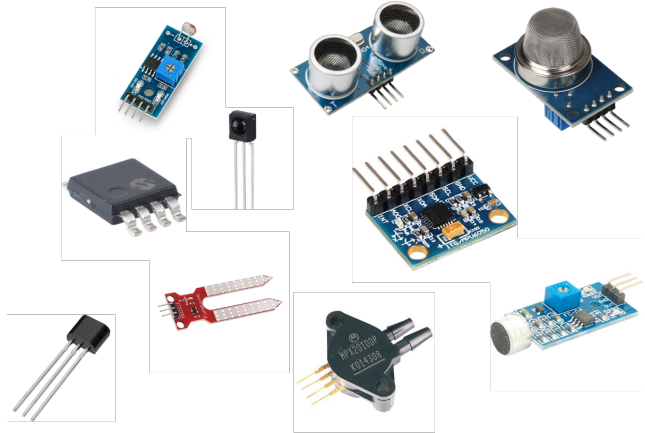


Fig.: sensors

The mean



$$\mu = \frac{1}{N} \sum_{n=1}^N V_n = \frac{1}{5} (5 + 5 + 10 + 10 + 10) = 8$$

The standarddeviation

	Player1	Player2	Player3	Player4	Player5	Mean
Team A	1.89m	2.10m	1.75m	1.98m	1.85m	1.914m
Team B	1.94m	1.90m	1.97m	1.89m	1.87m	1.914m

$$x_n - \mu = x_n - 1.914m$$

	Player1	Player2	Player3	Player4	Player5
Team A	-0.024m	0.186m	-0.164m	0.066m	-0.064m
Team B	0.026m	-0.014m	0.056m	-0.024m	-0.044m

The variance

	Player1	Player2	Player3	Player4	Player5
Team A	-0.024m	0.186m	-0.164m	0.066m	-0.064m
Team B	0.026m	-0.014m	0.056m	-0.024m	-0.044m

$$(x_n - \mu)^2 = (x_n - 1.914m)^2$$

	Player1	Player2	Player3	Player4	Player5
Team A	0.000576m ²	0.034596m ²	0.026896m ²	0.004356m ²	0.004096m ²
Team B	0.000676m ²	0.000196m ²	0.003136m ²	0.000576m ²	0.001936m ²

The variance

$$\sigma^2 = \frac{1}{N} \sum_{n=1}^N (x_{B_n} - \mu)^2$$

$$\sigma_A^2 = \frac{1}{N} \sum_{n=1}^N (x_{A_n} - \mu)^2 = \frac{1}{5} (0.000576 + 0.034596 + 0.026896 + 0.004356 + 0.004096) = 0.014m^2$$

$$\sigma_B^2 = \frac{1}{N} \sum_{n=1}^N (x_{B_n} - \mu)^2 = \frac{1}{5} (0.000676 + 0.000196 + 0.003136 + 0.000576 + 0.001936) = 0.0013m^2$$

- **Definition:** Covariance is a measure of how much two random variables change together. If the greater values of one variable mainly correspond with the greater values of the other variable, and the same holds for the lesser values (i.e., the variables tend to show similar behavior), the covariance is positive. In the opposite case, when the greater values of one variable mainly correspond to the lesser values of the other (i.e., the variables tend to show opposite behavior), the covariance is negative.

Mathematical Definition

Given two random variables X and Y , the covariance is defined as:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

where $\mathbb{E}[X]$ and $\mathbb{E}[Y]$ are the expected values (means) of X and Y , respectively.

The covariance

- **Example:** Consider two random variables, X and Y , representing the number of hours studied and the test scores of students. If higher hours of study tend to correspond with higher test scores, the covariance between X and Y would be positive.

Properties

- $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- $\text{Cov}(X, X) = \text{Var}(X)$ (Variance of X)
- If X and Y are independent, $\text{Cov}(X, Y) = 0$

Prediction

1. State prediction: $\hat{x}_k = A\hat{x}_{k-1} + Bu_{k-1}$
2. Covariance prediction: $P_k = AP_{k-1}A^T + Q$

Correction

3. Kalman Gain Prediction: $K_k = P_k H^T (H P_k H^T + R)^{-1}$
4. State Update: $\hat{x}_k = \hat{x}_k + K_k(z_k - H\hat{x}_k)$
5. Covariance Update: $P_k = (I - K_k H)P_k$

Kalman explained

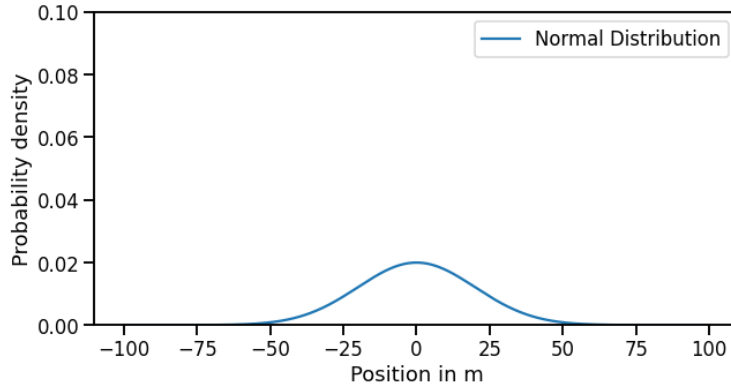


Fig.: Start Position at t_0

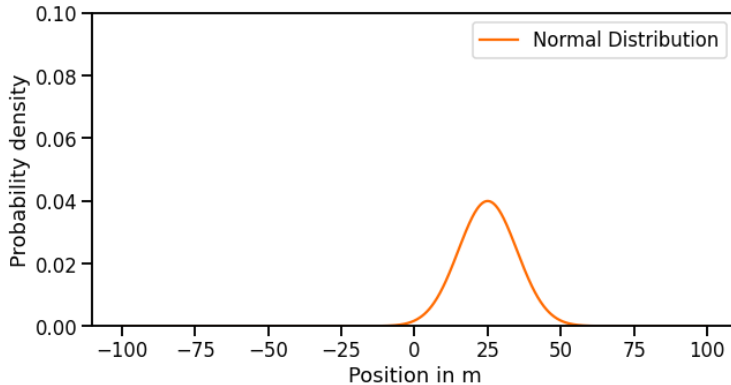


Fig.: Calculated Position at t_1

Kalman explained

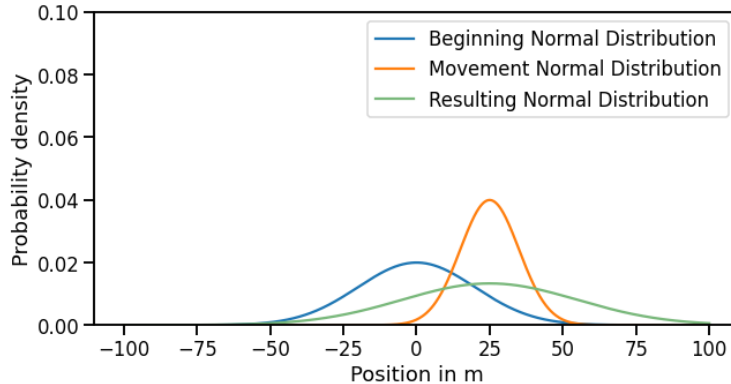


Fig.: Predicted Position at t_1

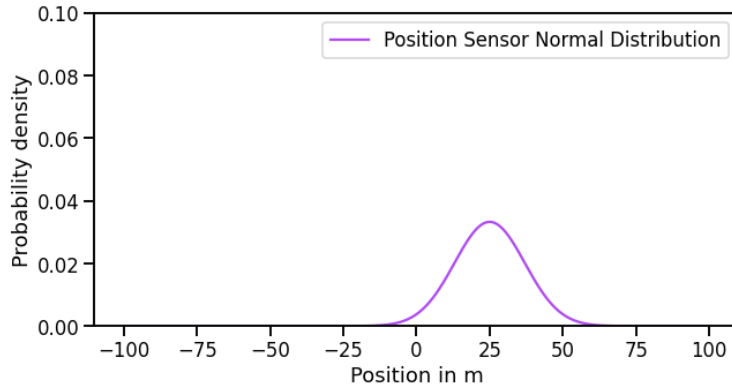


Fig.: Messured Sensor Data of Postion at t_1

Kalman explained

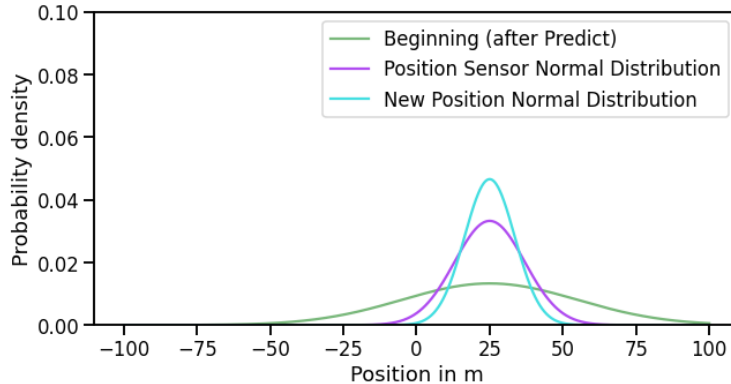


Fig.: Correction of the Kalman Gain after measurement at t_1

Takeaways

- The precision decreases with the prediction
- The precision increases with the correction

State Representation

We define the state vector x to include the position and velocity of the mouse in both x and y directions

$$x = \begin{bmatrix} x \\ y \\ v_x \\ v_y \end{bmatrix}$$

where:

- x is the position on the x-axis
- y is the position on the y-axis
- v_x is the velocity in the x-direction
- v_y is the velocity in the y-direction

Formulate the Transition Model

The state transition model describes how the state evolves from one time step to the next. If we assume a constant velocity model, the state transition can be expressed as:

$$x_{k+1} = Ax_k + w_k$$

where w_k represents the process noise, assumed to be Gaussian with zero mean and covariance Q .

Create the Transition Matrix

The transition matrix A for a constant velocity Model is:

$$A = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Δt describes the changes between two measurements.

The observation model relates the state to the measurements:

$$z_k = Hx_k + v_k$$

where v_k represents the measurement noise, assumed to be Gaussian with zero mean and covariance R .

The observation matrix H for direct measurement of position is:

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Prediction Step

The prediction step estimates the next state and its uncertainty:

- Predict the state:

$$x_{k+1} = Ax_k + Bu_k \quad (1)$$

- Predict the error covariance:

$$P_{k+1} = AP_kA^T + Q \quad (2)$$

Correction Step

The correction step updates the state estimate with the new measurement:

- Compute the Kalman gain:

$$K_k = P_k H^T (H P_k H^T + R)^{-1} \quad (3)$$

- Update the state estimate:

$$x_k = x_k + K_k (z_k - H x_k) \quad (4)$$

- Update the error covariance:

$$P_k = (I - K_k H) P_k \quad (5)$$

Kalman Filter

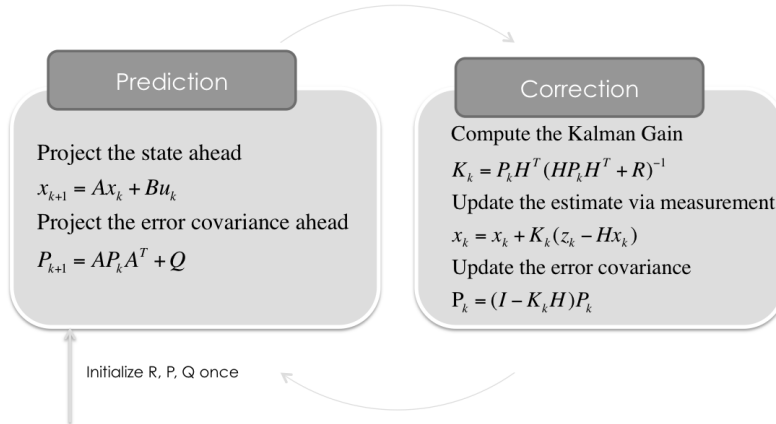


Fig.: Iterative Nature of the Kalman filter

Cheat Sheet - Math Symbols

Symbol	Meaning
k	Interval or iteration of the Kalman filter
z_k	Measurement vector. This contains the real-world measurement we received in this time step.
x_k	Newest estimate of the current "true" state.
P_k	Newest estimate of the average error for each part of the state.
R	Estimated measurement error covariance. Finding precise values for Q and R are beyond the scope of this guide.
Q	Estimated process error covariance. Finding precise values for Q and R are beyond the scope of this guide.
A	State transition matrix. Basically, multiply state by this and add control factors, and you get a prediction of the state for the next time step.
H	Observation matrix. Multiply a state vector by H to translate it to a measurement vector.
B	Control matrix. This is used to define linear equations for any control factors.
u_{k-1}	Control vector. This indicates the magnitude of any control system's or user's control on the situation.

TODO: Integration into a concrete use case

Messwerte des Beispiel

Imagine the following scenario: We are driving to the THWS, and are passing an funnel on our way. We are traveling on an average of $22.22m/s$ (80km/h).

Time in s	1	2	3	4	5	6	7	8	9	10
Distance in m	24.0	21.4	22.5	21.9	21.6	21.9	22.0	22.6	19.2	20.0

Example, 0. Iteration, Setup

Prediction

Project the state ahead

$$\hat{x}_k = 0$$

Project the error covariance ahead

$$P_k = 1$$

Correction

Compute the Kalman Gain

$$K_k = \frac{P_k}{P_k + R} = \frac{1}{1 + 0.58} = 0.6329$$

Update the estimate via Measurement

$$\begin{aligned}\hat{x}_{k+1} &= \hat{x}_k + K_k(z_k - \hat{x}_k) \\ &= 0 + 0.6329 \cdot (24.0 - 0) \\ &= 15.2\end{aligned}$$

Update the error covariance

$$\begin{aligned}P_{k+1} &= (I - K_k)P_k \\ &= (1 - 0.6329) \cdot 1 \\ &= 0.3671\end{aligned}$$

Example, 1. Iteration

Prediction

Project the state ahead

$$\hat{x}_k = 15.2$$

Project the error covariance ahead

$$P_k = 0.3671$$

Correction

Compute the Kalman Gain

$$K_k = \frac{P_k}{P_k + R} = \frac{0.3671}{0.3671 + 0.58} = 0.3876$$

Update the estimate via Measurement

$$\begin{aligned}\hat{x}_{k+1} &= \hat{x}_k + K_k(z_k - \hat{x}_k) \\ &= 15.2 + 0.3876 \cdot (21.4 - 15.2) \\ &= 17.6\end{aligned}$$

Update the error covariance

$$\begin{aligned}P_{k+1} &= (I - K_k)P_k \\ &= (1 - 0.3876) \cdot 0.3671 \\ &= 0.2248\end{aligned}$$

Example, 2. Iteration

Prediction

Project the state ahead

$$\hat{x}_k = 17.6$$

Project the error covariance ahead

$$P_k = 0.2248$$

Correction

Compute the Kalman Gain

$$K_k = \frac{P_k}{P_k + R} = \frac{0.2248}{0.2248 + 0.58} = 0.2793$$

Update the estimate via Measurement

$$\begin{aligned}\hat{x}_{k+1} &= \hat{x}_k + K_k(z_k - \hat{x}_k) \\ &= 17.6 + 0.2793 \cdot (22.5 - 17.6) \\ &= 19.0\end{aligned}$$

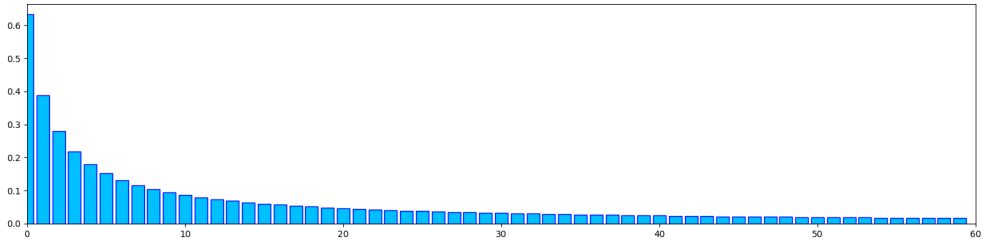
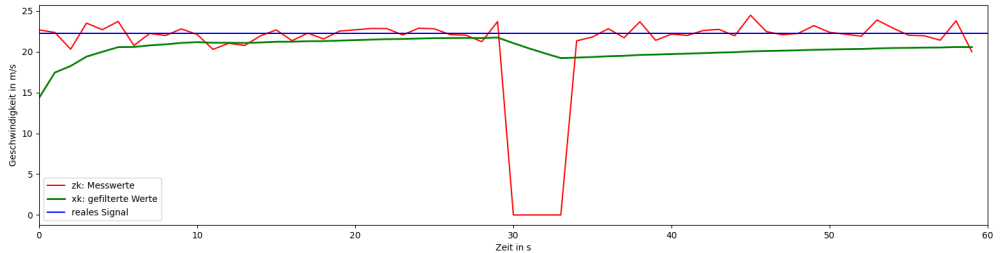
Update the error covariance

$$\begin{aligned}P_{k+1} &= (I - K_k)P_k \\ &= (1 - 0.2793) \cdot 0.2248 \\ &= 0.1620\end{aligned}$$

Iteration 1 to 10

k	0	1	2	3	4	5	6	7	8	9
z_k	24.0	21.4	22.5	21.9	21.6	21.9	22.0	22.6	19.2	20.0
K_k	0.63	0.38	0.27	0.21	0.17	0.15	0.13	0.11	0.10	0.09
P_k	1	0.36	0.22	0.16	0.12	0.10	0.08	0.07	0.06	0.06
$\hat{x}_k$	0	15.2	17.6	19.0	19.6	20.0	20.3	20.5	20.7	20.6

Tunnel



Summary



WELCOME TO
THE MATRIX!!!!!!

List of figures
